Movie Revenue Analysis

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Our Purpose

Understand patterns and relationships in the film industry

- Understand what drives success
- Analyze audience preferences
- Support decision-making for those who work in film

Data Overview

movie_id:	Unique identifier for each movie.
title:	Title of the movie.
vote_average:	Average rating of the movie (e.g., from 0 to 10).
vote_count:	Total number of votes the movie has received.
status:	Status of the movie (e.g., "Released", "Planned").
release_date:	Release date of the movie.
revenue:	Total revenue generated by the movie.
adult:	Indicates if movie is for adults (true/false)
budget:	Budget for movie production.
imbd_id:	Unique identifier for the movie on IMDb.
original_language:	Language in which movie was originally produced:
original _title:	Original title of the movie (if different from the main title).
overview:	Brief summary or description of the movie
popularity:	Popularity score of the movie, often based on various factors.
genres:	Comma-separated list of genres associated with the movie.
production_companies:	Comma-separated list of companies that produced the movie.
production_countries:	Comma-separated list of countries where the movie was produced.

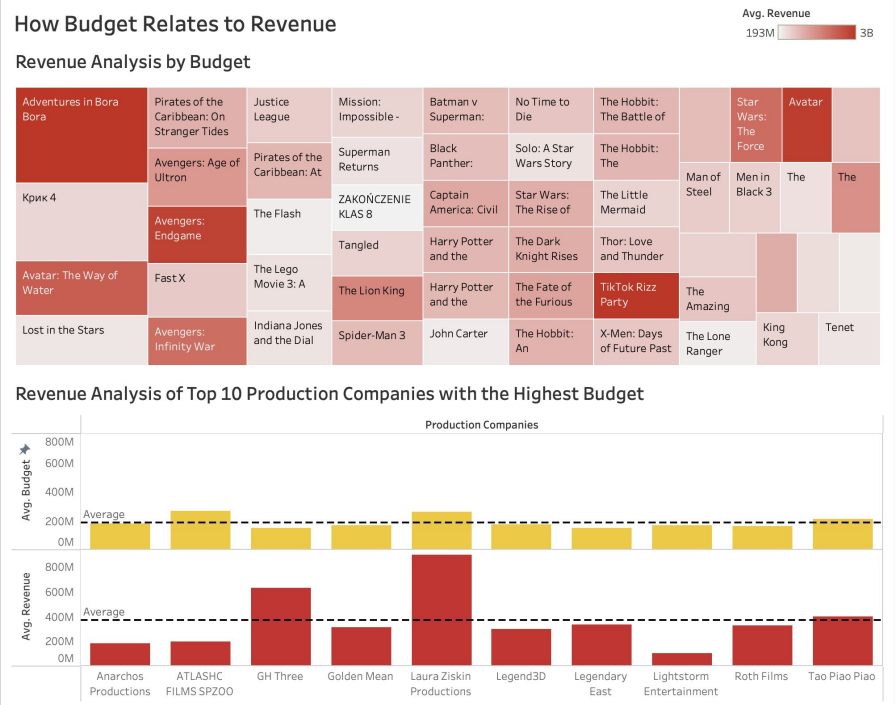
Spoken_languages: Comma-separated list of languages spoken in the movie.

Comma-separated list of languages spoken in the movie.

Data Overview

movie_id	title	vote_average	vote_count	status	release_date	revenue	adult	budget	imdb_id	original_lang	original_title	overview	popularity	genres	production_c	production_c	spoken_languages
4520010	Inception	8.364	34495	Released	7/15/10	825532764	FALSE	160000000	tt1375666	en	Inception	Cobb, a skille	83.952	Action, Scien	Legendary Pi	United Kingd	English, French, Japanese, Swahili
4520011	Interstellar	8.417	32571	Released	11/5/14	701729206	FALSE	165000000	tt0816692	en	Interstellar	The adventur	140.241	Adventure, D	Legendary Pi	United Kingd	English
4520012	The Dark Knig	8.512	30619	Released	7/16/08	1004558444	FALSE	185000000	tt0468569	en	The Dark Knig	Batman raise	130.643	Drama, Actio	DC Comics, I	United Kingd	English, Mandarin
4520013	Avatar	7.573	29815	Released	12/15/09	2923706026	FALSE	237000000	tt0499549	en	Avatar	In the 22nd c	79.932	Action, Adver	Dune Enterta	United States	English, Spanish
4520014	The Avengers	7.71	29166	Released	4/25/12	1518815515	FALSE	220000000	tt0848228	en	The Avengers	When an une	98.082	Science Ficti	Marvel Studi	United States	English, Hindi, Russian
4520015	Deadpool	7.606	28894	Released	2/9/16	783100000	FALSE	58000000	tt1431045	en	Deadpool	The origin sto	72.735	Action, Adver	20th Century	United States	English
4520016	Avengers: Infi	8.255	27713	Released	4/25/18	2052415039	FALSE	300000000	tt4154756	en	Avengers: Infi	As the Avenge	154.34	Adventure, A	Marvel Studi	United States	English, Xhosa
4520017	Fight Club	8.438	27238	Released	10/15/99	100853753	FALSE	63000000	tt0137523	en	Fight Club	A ticking-time	69.498	Drama	Regency Ente	United States	English
4520018	Guardians of	7.906	26638	Released	7/30/14	772776600	FALSE	170000000	tt2015381	en	Guardians of	Light years fr	33.255	Action, Scien	Marvel Studi	United States	English

Dashboard 1



- Tree Map:
  - Revenue → Color
  - Budget → Size
  - Marks → Square

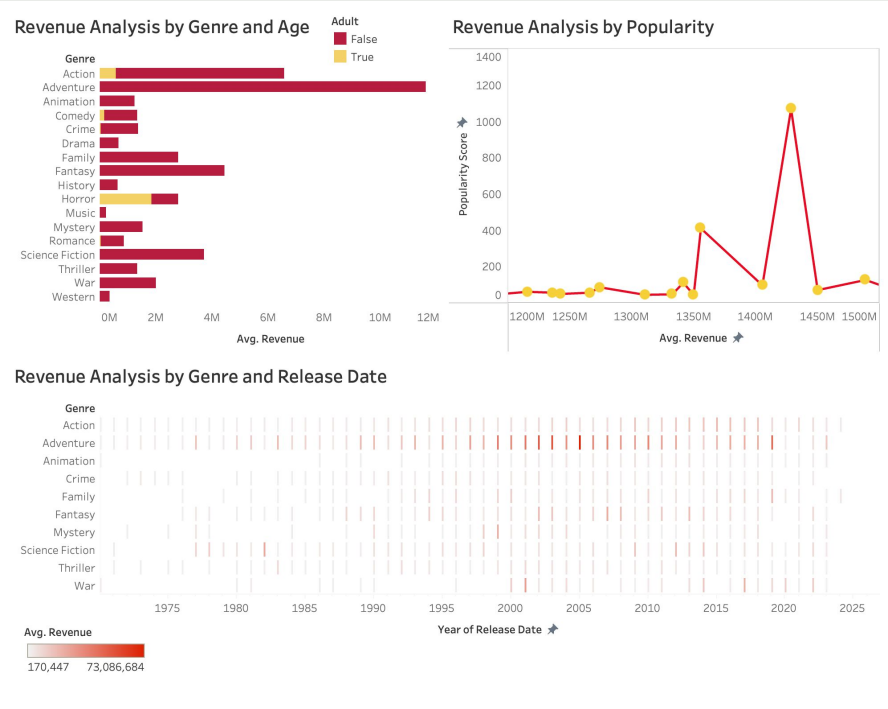
The bigger and more pigmented the square the higher the revenue/budget

- Bar Chart:
  - Created calculated field to split genre strings
  - Production companies → columns
  - Budget and revenue → rows
  - Added Average dotted line

Findings

- Higher budget leads to high revenue production
- Not many films over sell their budget costs
- Top 10 production companies often make three times as much of their revenue on average

Dashboard 2

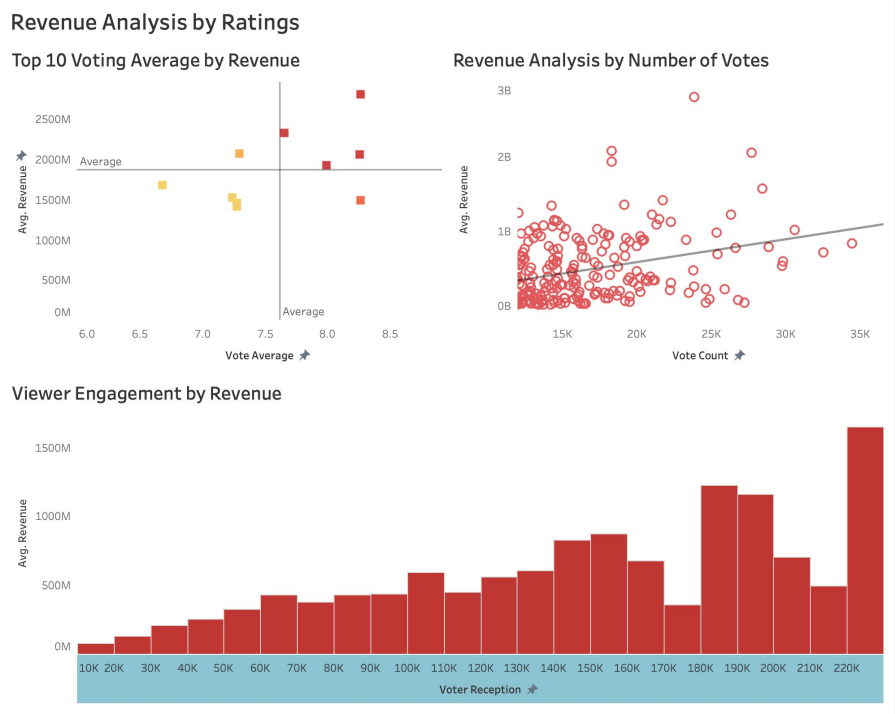


- Stacked Bar-Chart:
  - Genre → Rows
  - Revenue → Columns
  - Adults → Color
- Dual-Axis:
  - Revenue → columns
  - Popularity → rows x2 (one as a line chart another with circle chart)
  - Select dual axis
- Heat Map:
  - Release date → columns (selected year)
  - Genre → rows
  - Revenue → colors

Findings

- Adventure, action, and fantasy are the three genres with the highest grossings
- Films that are for adults and children are much more profitable
- Rise in popularity in film in more recent years
- Popularity score is related to increase in revenue

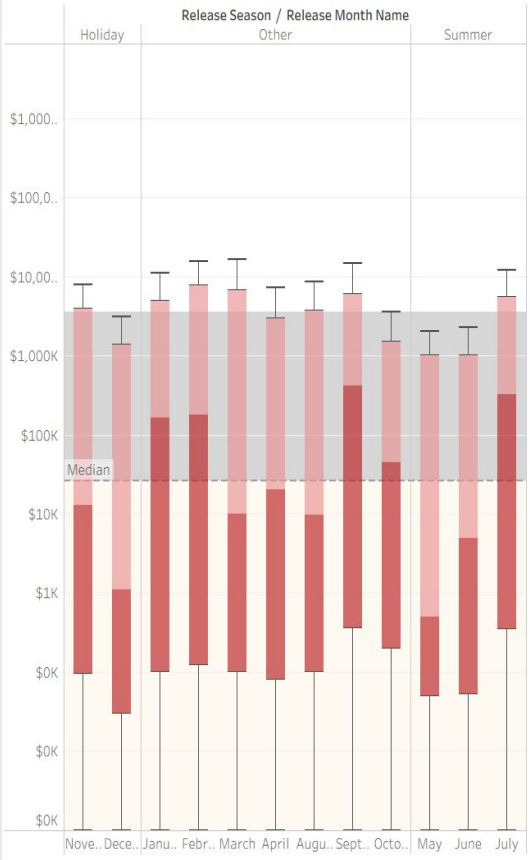
Dashboard 3



- Clustering:
  - Filtered top 10 revenues
  - Voting avg. →columns
  - Revenue avg. → rows
  - Added two average lines
  - Clustered by quadrants
- Scatter plot with a trend line:
  - Vote count → columns
  - Avg. revenue → rows
  - Filtered vote count
- Bar Chart:
  - Created a calculated field called “voter reception” to analyze viewer engagement (vote count x vote score)
  - Voter reception →columns
  - Avg. revenue → rows

Findings

- Movies with higher voting averages produce higher revenue
- Number of votes is statistically significant to avg. revenue (p-value= <0.0001)
- Higher viewer engagement leads to higher revenue



Seasonal Revenue Analysis

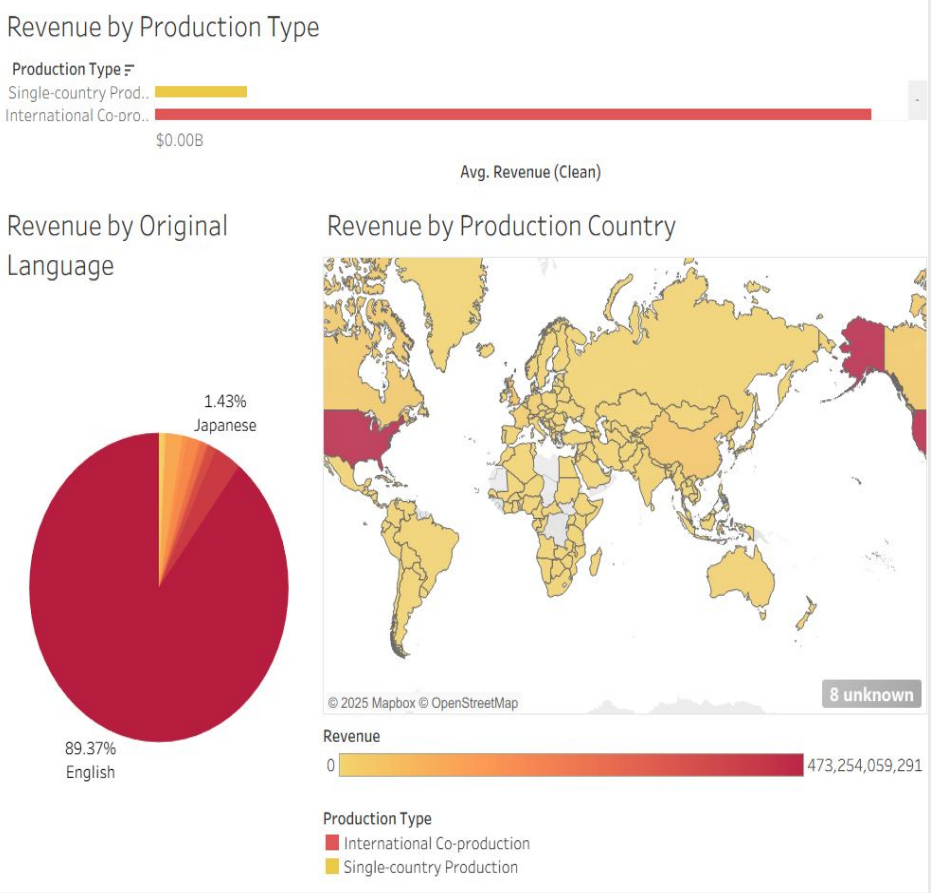
- Revenue distribution plotted by release month, grouped into Holiday, Summer, and Other
- Used box plots to capture spread, median, and outliers
- Added overall median reference line for comparison across months

Revenue Trend Over Time

- Release Year (binned) plotted against moving average of average revenue
- Applied a smoothed trend line to reduce volatility and highlight long-term patterns

Findings

- Movies released during holiday and peak summer months tend to generate higher median revenues compared to off-season releases.
- Revenue growth shows a strong upward trend from the late 1980s through early 2000s, indicating industry expansion and higher box-office potential over time.
- Recent years show greater volatility and slight decline, suggesting market saturation and increased competition from alternative distribution platforms.



Revenue by Production Type

- Compared average revenue across single-country productions and international co-productions
- Used a horizontal bar chart for direct revenue comparison

Revenue by Original Language

- Visualized revenue share by original language using a pie chart
- Highlighted dominance of English-language films relative to other languages

Revenue by Production Country

- Displayed total revenue by production country using a world heat map
- Color intensity represents cumulative box office performance across regions

Findings

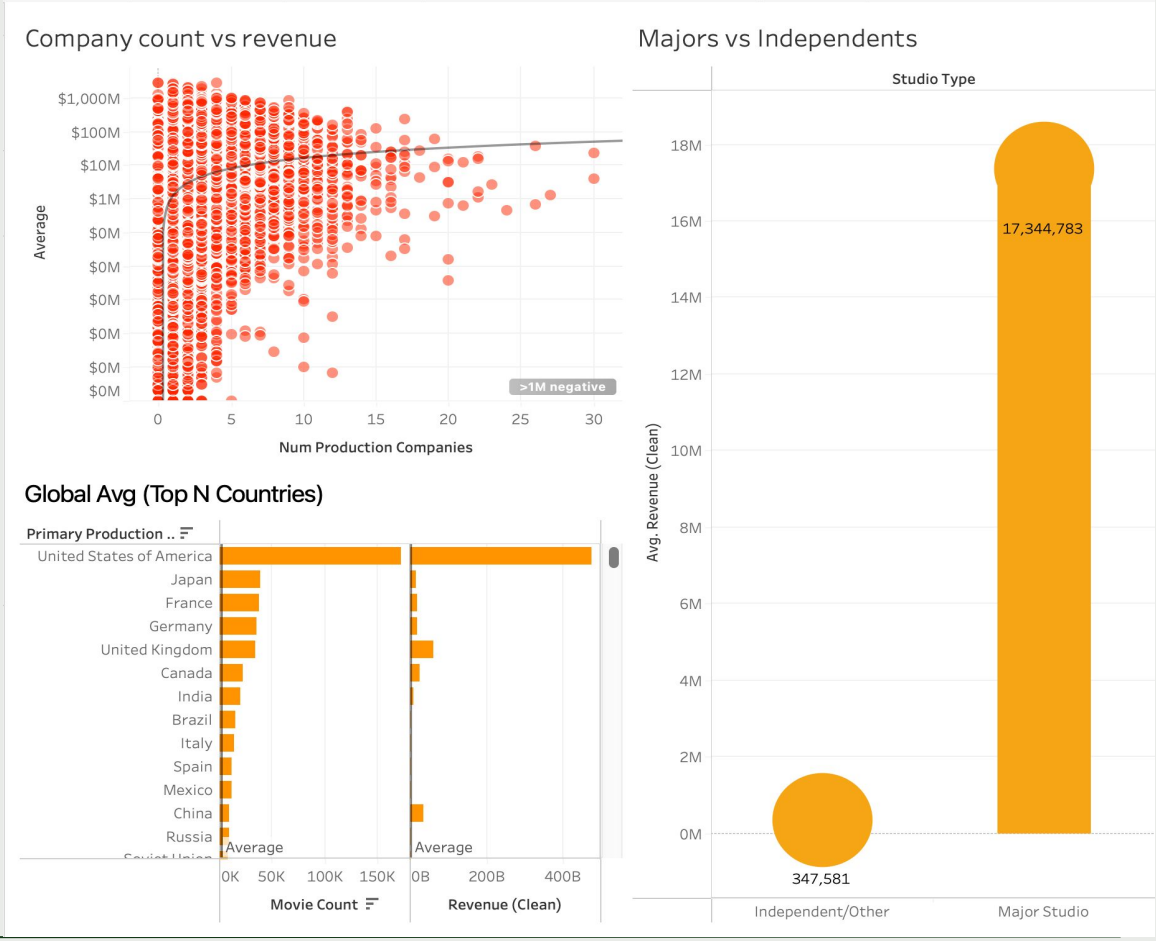
- International co-productions generate significantly higher average revenue than single-country productions, indicating benefits from cross-border financing and market access.
- English-language films dominate global box office revenue, accounting for the majority of total revenue compared to non-English productions.
- Box office revenue is highly concentrated in a small number of countries, with the United States and select international markets contributing disproportionately to global earnings.

Dashboard 4

Company Count Vs revenue

Majors vs independents

Global Average of Top N countries



Findings

It demonstrates that **revenue is not evenly distributed across the film industry**. Instead, it is concentrated among:

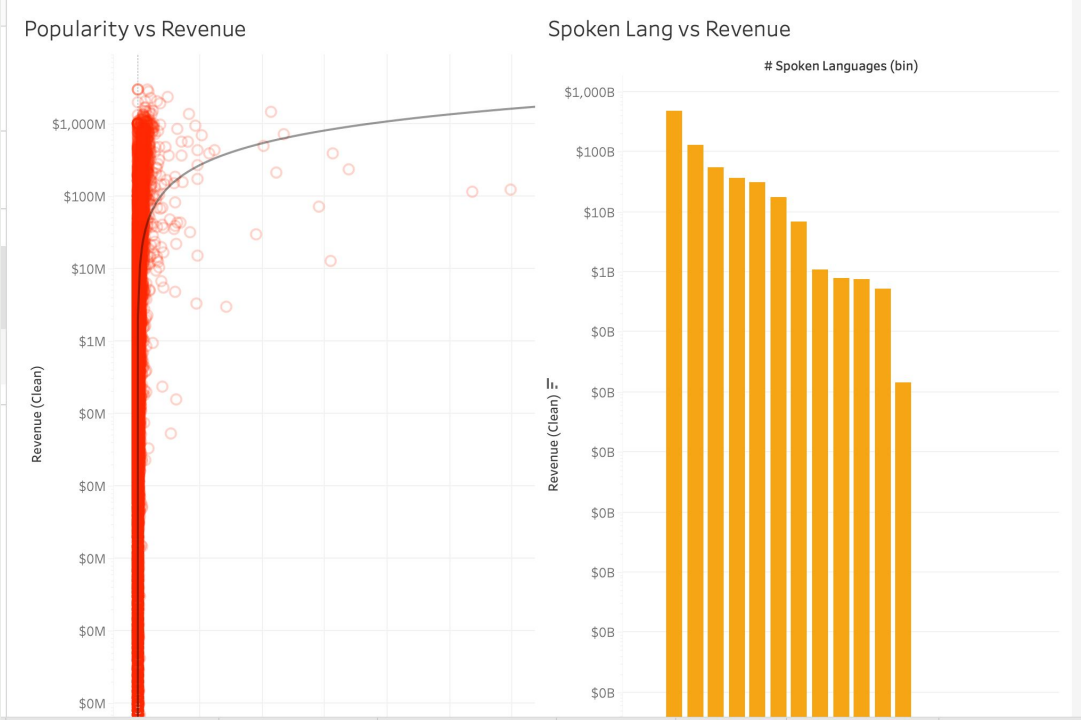
- **major studios**,
- **high-collaboration productions**, and
- **U.S.-centric production ecosystems**.

This structural imbalance helps explain why a small subset of films captures a disproportionate share of global box office revenue.

Dashboard 5

Popularity vs Revenue

Spoken languages vs revenue



Findings

It shows that demand signals matter more than breadth signals:

- Popularity is a strong and consistent predictor of higher revenue.
- Language diversification alone does not translate into commercial success and may dilute focus rather than expand market reach.

Together, these results indicate that effective marketing and audience anticipation outweigh content localization in driving box office outcomes.

Conclusion

- Movie box office performance is driven by a combination of budget, release timing, production strategy, and audience engagement, rather than any single factor.
- Higher budgets generally lead to higher revenues, but profitability varies widely, indicating that budget alone is not a reliable predictor of success.
- Movies released during summer and holiday seasons consistently earn more revenue, highlighting the importance of strategic release timing.
- International co-productions, English-language films, and productions from major markets tend to dominate global revenue, reflecting the value of broader market reach.
- Audience engagement—measured through ratings, vote counts, popularity, and overall reception—shows a strong positive relationship with revenue.
- Overall, successful films align financial investment, global accessibility, and strong audience response, reinforcing the need for data-driven decision-making in the movie industry.

Sources

https://www.kaggle.com/datasets/mohammedalsubaie/movies?resource=download

Tableau Link

https://public.tableau.com/app/profile/shrey.soni7744/viz/shared/K686Z563Y

Thank You!